

Jamie Seoh

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EDUCATION

University of Waterloo

Bachelor of Software Engineering

Sep 2025 – Present

3.95/4.00 (91.18%) GPA

TECHNICAL SKILLS

Languages: Python, C/C++, Bash, TypeScript & JavaScript, HTML & CSS, SQL, Java

Frameworks and Libraries: PyTorch, NumPy, React, Next.js, Svelte, FastAPI, Flask, Tailwind CSS

Cloud and DevOps: Docker, GCP, Azure, AWS, MongoDB, PostgreSQL, Git/GitHub

Embedded Systems: KiCad, Fusion 360, 3D Printing, Figma, Teensy, ESP32, Breadboarding, Arduino

EXPERIENCE

AI Software Engineer Intern

May 2026 – Aug 2026

Cerebras Systems | C++, Python, Assembly, Bash, GTest

Toronto, Ontario

- Cut production fault-triage time **30%** by pinpointing faulty compute units across parallel inference workloads.
- Expanded **hardware fault injection** with crash, stall, and slowdown modes, implementing end-to-end support to the inference pipeline from the runtime to on-device kernel assembly with automated testing.
- Reduced false-positive on-call alerts by redesigning faulty-hardware detection with a two-tier alerting system.
- Developed **KV-cache** debug tooling to emulate stalls, enabling diagnosis of difficult-to-isolate hardware failures.
- Built developer tooling: a static linter for a custom hardware assembly language and an observability dashboard.

Front-End Developer

Sep 2023 – Apr 2024

SolveXchange | HTML, CSS, JavaScript, Flask, SQLAlchemy

Hybrid

- Shipped and maintained responsive frontend interfaces serving **100+** active users for a non-profit platform.
- Built secure authentication and data workflows with **Flask**, **SQLAlchemy**, and **RESTful** APIs.

EXTRACURRICULAR INVOLVEMENT

Logistics Organizer @Hack the North

Feb 2026 – Present

- Organized **Hack the North**, Canada's largest hackathon with **1,000+** participants and **55+** sponsors.
- Led all volunteer operations end-to-end, managing **300+** volunteers across task coordination and execution.
- Oversaw **300+** 3D-printed keychains from design to production given time and material constraints.

PROJECTS

Self-Playing MIDI Keyboard | C++, Arduino, Teensy 4.0, Hall-effect Sensors, Solenoids, 3D Printing

- Designed a 25-key MIDI keyboard with 3D-printed keys, velocity sensing, and USB-MIDI/synthesizer modes.
- Engineered a **Hall-effect sensing system** using multiplexed analog inputs, reducing pin usage from 25 to 6.
- Developed Teensy firmware for sensor processing, MIDI handling, velocity detection, and real-time key control.
- Built a **self-playing system** using solenoids, MOSFET switching, and PWM for independent key actuation.

Quota — DeltaHacks12 First Place 🏆 | TypeScript, Next.js, React Flow, Gemini, LangChain, MongoDB

- Built VS Code extension with **TypeScript** for inline cost annotations, heatmaps, and one-click optimizations.
- Indexed codebases **<3s** for large codebases with **10,000+** lines using **AST parsing** to detect inefficiencies.
- Delivered real-time API cost insights via regex and usage caching, cutting API spend by up to **40%**.
- Created **React Flow** sandbox for budget-aware system planning with **RAG Gemini** chatbot by LangChain.

Nebula AI 🏆 | React Native, FastAPI, PostgreSQL, Docker, GCP

- Developed a mobile app that turns handwritten notes into a searchable, **LaTeX**-formatted AI knowledge base.
- Enabled semantic search over **1,536-dim embeddings**, cutting retrieval time by **40%** with IVFFlat indexing.
- Implemented an **OCR+AI pipeline** with Gemini/OpenAI, enabling handwritten text extraction and summaries.
- Containerized backend with **Docker**, deployed to **Google Cloud Run** with automated CI/CD pipelines.

Personal CRM 🏆 | Next.js, FastAPI, PostgreSQL, TailwindCSS

- Built a robust full-stack CRM to centralize and prioritize personal & professional relationships at scale.
- Architected backend with **20+ REST** endpoints, enabling pagination & dynamic fetching for growing datasets.
- Improved system reliability through caching layers and rate limiting, reducing server load by **30%**.